

Salt, volume and the prevention of hypertension.



The evidence supporting the thesis that hypertension can be prevented by eliminating salt from the diet is based on four principal sources: (1) epidemiological studies in unacculturated peoples showing that the prevalence of hypertension is inversely correlated with the degree of salt intake; (2) hemodynamic studies suggesting that the development of chronic experimental hypertension is a homeostatic response to a maintained increase in extracellular fluid volume (ECF); (3) evidence that the ECF of "salt eaters" is expanded in comparison to that of "no-salt eaters"; and (4) investigations in hypertensive patients receiving either diets greatly restricted in salt or continuous diuretic therapy which correlate the fall in blood pressure with a reduction in ECF. Although this mechanism of essential hypertension is still obscure the evidence is very good if not conclusive that reduction of salt in the diet to below 2 g/day would result in the prevention of essential hypertension and its disappearance as a major public health problem.